

MRINAL MISHRA, Ph.D.

BIOINFORMATICS ANALYST | COMPUTATIONAL GENOMICS | MACHINE LEARNING FOR PHARMACOGENOMICS

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Bioinformatics Analyst and computational biologist with a Ph.D. in Biology and hands-on experience building genomic analysis workflows, machine learning models, and multi-omics studies. Focused on pharmacogenomics, gene expression prediction, variant interpretation, population genomics, and research that improves representation of historically excluded populations in precision medicine.

TOP PICKS

Machine Learning Built deep learning and regression workflows using TensorFlow, Keras, PyTorch, and scikit-learn for gene expression and biological prediction problems.	Genomics Led whole-genome sequencing, variant calling, functional annotation, eQTL, gene enrichment, and population genomics analyses.
Multi-omics Published work connecting genomics, transcriptomics, and pharmacogenomics in African American populations.	Research Output Author/co-author on peer-reviewed publications across pharmacogenomics, molecular ecology, metagenomics, and computational biology.

FEATURED EXPERIENCE

Bioinformatics Analyst

Northwestern University, Perera Lab, Feinberg School of Medicine | 2022 - Present

Develop machine learning and genomics workflows for pharmacogenomics research, with emphasis on African American cohorts and gene expression regulation.

- Developing deep learning models to predict gene expression across the human genome.
- Analyzing drug treatment effects in hepatic cells from African American samples.
- Contributed to peer-reviewed research on drug perturbation, eQTL mapping, and multi-omics equity in precision medicine.

Ph.D. Candidate, Computational Biology

University of Florida, Department of Biology | 2015 - 2021

Executed whole-genome sequencing analysis, functional annotation, gene enrichment, variant detection, and population genomics research.

- Researched hybridization of four-eyed sleeper fish populations across regions of the East China Sea.
- Identified, evaluated, and categorized population variants for dissertation and publication work.
- Created R workflows for population genomics and developed a Python package to filter low-quality variants.

Graduate Teaching Assistant

University of Florida, Department of Biology | 2015 - 2021

Supported undergraduate instruction in genetics, genomics, bioinformatics, molecular biology, and lab techniques.

- Mentored students in individual and group settings on coursework and project work.
- Designed and delivered bioinformatics workshops for undergraduate students.
- Taught concepts including DNA/RNA structure, gene expression, proteomics, PCR, gel electrophoresis, and fermentation.

Research Trainee

VTT Technical Research Centre, Turku, Finland | 2012 - 2014

Applied statistical modeling and image analysis to study cellular heterogeneity in 3D cancer tumor-cell images.

- Developed a cell image analysis pipeline covering acquisition, quality control, cleanup, exploratory analysis, and hypothesis-driven analysis.
- Wrote R scripts to segment cancer cell images and extract morphological feature sets.
- Prepared raw data for statistical analysis and machine learning models.

SELECTED CREDITS

Summer Research Trainee

Vellore Institute of Technology, Tamil Nadu, India | 2010

Conducted in silico pathway analysis of pancreatic cancer genes using computational biology tools.

- Integrated expression-profile data into molecular interaction networks using Cytoscape.
- Identified interactions among KRAS, CDKN2A, MADH4, TP53, ARMET, and neighboring genes.
- Presented findings in a poster on computational analysis of pancreatic cancer genetic networks.

EDUCATION

Ph.D. in Biology | University of Florida, Gainesville, FL | 2021

M.S. in Bioinformatics | University of Turku, Turku, Finland | 2015

Bachelor of Technology, Bioinformatics | Vellore Institute of Technology, Vellore, India | 2012

TECHNICAL STACK

Programming: R, Python, Biopython

ML: TensorFlow, Keras, PyTorch, scikit-learn, regression, convolutional neural networks, autoencoders

GWAS / Sequencing: GATK, BWA, Bowtie, DESeq2, BayeScan, KAT, variant calling, eQTL mapping

Annotation: MAKER, Blast2GO

Population Genetics: mothur, FSTAT, Genepop, STRUCTURE, GenAIEx, POPGENE, GDA, Micro-Checker, MrBayes, TNT, MEGA

PUBLICATIONS

- Yang G, Mishra M, Perera MA. Multi-Omics Studies in Historically Excluded Populations: The Road to Equity. Clin Pharmacol Ther. 2023;113(3):541-556. doi: 10.1002/cpt.2818.
- Zhong Y, De T, Mishra M, Avitia J, Alarcon C, Perera MA. Leveraging drug perturbation to reveal genetic regulators of hepatic gene expression in African Americans. Am J Hum Genet. 2023;110(1):58-70. doi: 10.1016/j.ajhg.2022.12.005.
- Liu Q, Mishra M, Saxena AS, Wu H, Qiu Y, Zhang X, You X, Ding S, Miyamoto MM. Balancing selection maintains ancient polymorphisms at conserved enhancers for olfactory receptor genes of a Chinese marine fish. Molecular Ecology. 2021;30(16):4023-4038.
- Li H, Mishra M, Ding S, Miyamoto MM. Diversity and Dynamics of Candidatus Endobugula and Other Symbiotic Bacteria in Chinese Populations of the Bryozoan, Bugula neritina. Microbial Ecology. 2019;77(1):243-256.
- Ding S, Mishra M, Wu H, Liang S, Miyamoto MM. Characterization of hybridization within a secondary contact region of Bostrychus sinensis in the East China Sea. Heredity. 2018;120(1):51-62.
- Mishra M, Liang S, Ke C, Ding S, Miyamoto MM. The Potential Utility of Microsatellite Markers for Resolving the Phylogeographic Structure of the Rock Shell in the Northwest Pacific. J Phylogenetics Evol Biol. 2016;4:4.
- Gupta R, Mishra M, Tandon P, Hisaria R. Novel mutation predicted to disrupt SGOL1 protein function. Egyptian Journal of Medical Human Genetics. 2013;14:149-155.
- Mishra M, Kumar A. Computational analysis of genetic network involved in pancreatic cancer in human. BMC Bioinformatics. 2011;12 Suppl 11:A11.

CERTIFICATIONS

- Supervised Machine Learning: Regression and Classification
- Advanced Learning Algorithms
- Python for Genomic Data Science
- Genomic Data Science with Galaxy
- Introduction to Genomic Technologies

AWARDS & SERVICE

- Grinter Fellowship, University of Florida, 2015
- Travel Fellowship Award, Swiss Institute of Bioinformatics, 2013
- Best Poster Award, International Society for Computational Biology, 2011
- Conference Volunteer, ISMB/ECCB Berlin Conference, 2013
- Conference Volunteer, ISMB/ECCB Vienna Conference, 2011

CODE PORTFOLIO

- GitHub portfolio includes repositories for image segmentation and feature extraction, VCF filtering, QSAR prediction modeling, genome-read matching, and R-based file handling / plotting functions.
- Public profile: github.com/mrimis